

Finite Dyadic Sinc Products and Piecewise-Polynomial Approximants to Rvachev’s Up-Function

Exact splines, sharp convergence laws, Bell–Bernoulli expansions,
and positive quadrature acceleration

Research report prepared from the `ProveIt` Fabius-function documentation
Computational supplement: `finite_sinc_experiments.py`

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Abstract

Let

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

under the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i x \xi} dx$. This report studies the inverse transforms p_n of the first n factors. They are nonstationary dyadic box splines: compactly supported, nonnegative, even, log-concave, piecewise polynomial of degree $n-1$, and C^{n-2} . Their knots unexpectedly form a uniform dyadic grid, while the jumps of the top derivative carry the signed Thue–Morse sequence. An explicit truncated-power formula, exact cell formulas, recursions, Fourier-zero multiplicities, partition-of-unity identities, cumulants, and sharp derivative geometry are obtained.

The main convergence result is an exact law, not merely an upper bound. For every $r \geq 0$ and $n \geq r+3$,

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The extremum occurs at an explicit dyadic point. The associated distribution functions satisfy the exact Kolmogorov law $d_K = 1/(9 \cdot 4^n)$. At all orders, p_n admits a uniform differential expansion whose coefficients are complete Bell polynomials in Bernoulli-number data from $1/\Phi$; geometric Richardson combinations then give signed 4^{-Mn} extrapolation with uniformly bounded coefficient variation.

A second, positive acceleration theory is developed from a new uniform-scale-mixture representation of up . The exact tail is a mixture of uniform boxes governed by a probability measure ν on $[0, 1]$. Positive Gauss, Radau, and Lobatto quadrature for ν turns the continuum tail into finite positive mixtures of boxes. A one-signed Peano-kernel argument and the dyadic derivative plateaux give exact C^r errors for the whole hierarchy. In particular, a single variance-matched extra box yields an exact 16^{-n} law, resolving a positive variance-matching direction left open in the repository’s frontier volume; right-Radau rules preserve the exact support $[-1, 1]$ and attain arbitrarily high algebraic orders. Reproducible FFT and exact-rational experiments verify the formulas to numerical precision.

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1 Scope, source audit, and novelty boundary

1.1 Documents examined

The source audit covered the living TeX documents under `Analysis/FabiusFunction/docs`, especially the primary synthesis *The Fabius Function and Rvachev’s Up-Function*, the mathematical glossary, the Lean walkthrough, and the consolidated volume *Non-formalized Research Frontiers*. The recursive archive index was also examined to identify the historical TeX families: early notes, standalone studies, primary-exposition drafts, supplements, q -calculus articles, Lean walkthroughs, glossaries, and per-topic frontier dossiers. The archive explicitly records frozen first-introduced versions and prunes byte-identical copies already present in the live tree, so the living consolidated sources are the appropriate mathematical baseline for avoiding duplication [1, 2, 3].

The existing corpus already develops the probability law of up, the infinite sinc product, exact dyadic arithmetic, Thue–Morse identities, derivative tilings, moments, q -binomial structure, endpoint Lambert– W asymptotics, and a repeated-integration spline f_n . In particular, the frontier volume proves a sharp error law for f_n and explicitly contrasts it with the ordinary prefix density p_n , whose sharp law is not studied there. It then asks for a positive compactly supported tail replacement with matching variance, predicting a possible 16^{-n} rate but asserting no such scheme [2].

1.2 What is new here

The principal additions of this report are:

1. the exact uniform-grid truncated-power representation of the ordinary inverse transform p_n , with Thue–Morse jump weights;
2. a sharp derivative-plateau theorem for p_n ;
3. the exact C^r prefix error law and exact Kolmogorov distance;
4. a uniform all-orders expansion of p_n in even derivatives of up, with Bell–Bernoulli coefficient formulas;
5. stable $q = 1/4$ Richardson weights and their closed q -binomial form;
6. a uniform-scale-mixture representation of the exact tail;
7. a general positive quadrature theorem giving exact error constants for Gauss, Radau, and Lobatto tail surrogates;
8. explicit positive fourth- and sixth-order schemes, including exact-support accelerants.

Every theorem below is accompanied by a mathematical proof. These results have not been added to the repository’s Lean development and should therefore be read as a research contribution awaiting independent review and formalization. Numerical experiments are used as verification, never as a substitute for proof.

2 Fourier normalization and the probabilistic head–tail model

2.1 The infinite product

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i x \xi} d\xi. \quad (1)$$

For $a > 0$, define the centered uniform density

$$u_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x). \quad (2)$$

Then

$$\widehat{u}_a(\xi) = \text{sinc}(2\pi a\xi). \quad (3)$$

Rvachev’s normalized up-function is the even C^∞ probability density on $[-1, 1]$ whose transform is

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right). \quad (4)$$

This normalization agrees with the compactly supported function studied by Rvachev and Arias de Reyna [7, 8].

2.2 Partial products and random sums

For $n \geq 1$, set

$$\Phi_n(\xi) = \prod_{j=0}^{n-1} \text{sinc}\left(\frac{\pi\xi}{2^j}\right), \quad p_n = \mathcal{F}^{-1}\Phi_n. \quad (5)$$

By Equation (3),

$$p_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}}. \quad (6)$$

Let $U_0, U_1, \dots$ be independent $\text{Unif}[-1, 1]$ variables and put

$$S_n = \sum_{j=0}^{n-1} 2^{-j-1} U_j, \quad X = \sum_{j=0}^{\infty} 2^{-j-1} U_j. \quad (7)$$

Then p_n is the density of S_n and up is the density of X . Introduce

$$\delta_n = 2^{-n}, \quad A_n = 1 - 2^{-n}. \quad (8)$$

The omitted tail is a scaled independent copy of the whole sum:

$$X \stackrel{d}{=} S_n + \delta_n X', \quad X' \stackrel{d}{=} X, \quad X' \text{ independent of } S_n. \quad (9)$$

Consequently,

$$\Phi(\xi) = \Phi_n(\xi) \Phi(\delta_n \xi), \quad \text{up} = p_n * \rho_{\delta_n}, \quad \rho_{\delta}(x) = \delta^{-1} \text{up}(x/\delta). \quad (10)$$

The identity Equation (10) is the engine behind both the exact error law and all-order acceleration.

3 Exact inverse transforms: dyadic box splines with Thue–Morse knots

3.1 Uniform knot formula

Write

$$\tau_k = (-1)^{s_2(k)}, \quad (11)$$

where $s_2(k)$ is the binary digit sum. For $m = 0$, the convention $y_+^0 = \mathbf{1}_{(0, \infty)}(y)$ is used away from knots.

Theorem 3.1 (Exact truncated-power formula). *For every $n \geq 1$,*

$$p_n(x) = \frac{2^{n(n-1)/2}}{(n-1)!} \sum_{k=0}^{2^n-1} \tau_k \left(x + A_n - \frac{k}{2^{n-1}} \right)_+^{n-1}. \quad (12)$$

The knots are the uniformly spaced points

$$t_{n,k} = -A_n + \frac{k}{2^{n-1}}, \quad 0 \leq k < 2^n. \quad (13)$$

In distributions,

$$D^n p_n = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} \tau_k \delta_{t_{n,k}}. \quad (14)$$

Proof. For one box,

$$Du_a = \frac{1}{2a}(\delta_{-a} - \delta_a).$$

Convolving the n derivatives in Equation (6) gives a signed atomic measure. Choose a binary vector $b = (b_0, \dots, b_{n-1}) \in \{0, 1\}^n$; its atom lies at

$$-A_n + \sum_{j=0}^{n-1} b_j 2^{-j} = -A_n + \frac{k}{2^{n-1}}, \quad k = \sum_{j=0}^{n-1} b_j 2^{n-1-j},$$

and its sign is $(-1)^{\sum b_j} = \tau_k$. The normalization is

$$\prod_{j=0}^{n-1} \frac{1}{2 \cdot 2^{-j-1}} = \prod_{j=0}^{n-1} 2^j = 2^{n(n-1)/2}.$$

This proves Equation (14). Integrating n times from $-\infty$, using compact support, gives Equation (12). $\square$

The equal spacing is special to the dyadic weights. The 2^n subset sums of $1, 1/2, \dots, 2^{1-n}$ fill a complete integer grid after multiplication by 2^{n-1} , while their parity signs are exactly Thue–Morse. Thus a nonstationary convolution of boxes of different widths becomes a uniform-knot spline with highly structured signed top derivative.

3.2 Cell formula and exact arithmetic

Set

$$y = 2^{n-1}(x + A_n). \quad (15)$$

Inside the support, Equation (12) becomes

$$p_n(x) = \frac{2^{-(n-1)(n-2)/2}}{(n-1)!} \sum_{0 \leq k \leq [y]} \tau_k (y - k)^{n-1}. \quad (16)$$

This has several immediate consequences.

Corollary 3.2 (Piecewise-polynomial and arithmetic structure). *The function p_n is supported on $[-A_n, A_n]$, is polynomial of degree at most $n-1$ on each of the 2^n-1 open knot cells, and belongs to $C^{n-2}(\mathbb{R})$ for $n \geq 2$. At rational arguments it has rational values; at dyadic arguments all computations may be performed with integers after one global power-of-two scaling.*

On the cell $(t_{n,J}, t_{n,J+1})$,

$$p_n^{(n-1)}(x) = 2^{n(n-1)/2} \sum_{k=0}^J \tau_k. \quad (17)$$

The elementary Thue–Morse prefix identity

$$\sum_{k=0}^J \tau_k = \begin{cases} \tau_{J/2}, & J \text{ even}, \\ 0, & J \text{ odd} \end{cases} \quad (18)$$

therefore shows that every second top-derivative cell is identically zero, while the remaining cells carry a compressed Thue–Morse pattern.

3.3 Convolution and scaling recursions

Appending the final box in Equation (6) gives

$$p_n(x) = 2^{n-1} \int_{x-2^{-n}}^{x+2^{-n}} p_{n-1}(t) dt, \quad (19)$$

$$p'_n(x) = 2^{n-1} (p_{n-1}(x + 2^{-n}) - p_{n-1}(x - 2^{-n})). \quad (20)$$

The product shell $\Phi_n(\xi) = \text{sinc}(\pi\xi)\Phi_{n-1}(\xi/2)$ yields the equivalent dilation recursion

$$p_n(x) = \int_{2x-1}^{2x+1} p_{n-1}(s) ds, \quad (21)$$

$$p'_n(x) = 2(p_{n-1}(2x+1) - p_{n-1}(2x-1)). \quad (22)$$

These formulas permit exact construction of all cell polynomials by repeated primitive and shift operations. They avoid the severe cancellation that can occur in the global truncated-power sum for large n .

4 Structural properties of the finite-product splines

4.1 Shape, plateau, and partition of unity

Proposition 4.1 (Basic geometry). *For every $n \geq 1$:*

- (i) $p_n \geq 0$, p_n is even, and $\int p_n = 1$;
- (ii) $\text{supp } p_n = [-A_n, A_n]$;
- (iii) p_n is log-concave and therefore unimodal;
- (iv) $p_n(x) = 1$ for $|x| \leq \delta_n$;
- (v) for $n \geq 2$,

$$\sum_{m \in \mathbb{Z}} p_n(x - m) = 1, \quad p_n(x) + p_n(1 - x) = 1 \quad (0 \leq x \leq 1). \quad (23)$$

Proof. The first three assertions follow from convolution of centered uniform densities; log-concavity is preserved by convolution. For the plateau, split off the largest box $u_{1/2}$. The remaining convolution is supported on $[-1/2 + \delta_n, 1/2 - \delta_n]$. If $|x| \leq \delta_n$, the interval $[x - 1/2, x + 1/2]$ contains that support, so convolution with $u_{1/2}$ integrates the entire remaining density and equals 1.

At every nonzero integer k , the factor $\text{sinc}(\pi\xi)$ gives $\Phi_n(k) = 0$, while $\Phi_n(0) = 1$. Poisson summation therefore gives the first identity in Equation (23). On $[0, 1]$, the support radius is less than 1, so only the translates $m = 0, 1$ survive; evenness converts $p_n(x - 1)$ to $p_n(1 - x)$. $\square$

4.2 Fourier zeros

Let $v_2(k)$ denote the exponent of 2 in a nonzero integer k .

Proposition 4.2 (Exact zero multiplicities). *For $k \in \mathbb{Z} \setminus \{0\}$,*

$$\text{ord}_{\xi=k} \Phi_n = 1 + \min\{v_2(k), n - 1\}. \quad (24)$$

For fixed k , this stabilizes to $1 + v_2(k)$, the multiplicity of the corresponding zero of Φ .

This 2-adic spectral stratification is the Fourier counterpart of the Thue–Morse jump pattern. It also explains why the splines reproduce constants by integer translation, while higher-order cancellation appears selectively on dyadic frequency sublattices.

4.3 Cumulants and the variance defect

Let B_{2m} be the Bernoulli numbers with $B_2 = 1/6$ and $B_4 = -1/30$. The even cumulants of X are

$$\kappa_{2m}(X) = \frac{B_{2m}}{2m(1 - 2^{-2m})}, \quad \kappa_{2m+1}(X) = 0. \quad (25)$$

Indeed, a uniform $U \sim \text{Unif}[-1, 1]$ has $\kappa_{2m}(U) = 2^{2m} B_{2m} / (2m)$, and the geometric weights sum exactly. From Equation (9),

$$\kappa_{2m}(S_n) = (1 - \delta_n^{2m}) \kappa_{2m}(X). \quad (26)$$

In particular,

$$\mathbb{E}X^2 = \frac{1}{9}, \quad \mathbb{E}X^4 = \frac{19}{675}, \quad \mathbb{E}X^6 = \frac{583}{59535}, \quad (27)$$

and

$$\text{Var } X - \text{Var } S_n = \frac{\delta_n^2}{9}. \quad (28)$$

The raw prefix's leading 4^{-n} error is exactly this missing variance transported through the second derivative.

4.4 Sharp derivative plateaux

Define

$$c_m = 1 - 2^{-m} \quad (m \geq 0), \quad c_0 = 0. \quad (29)$$

The living primary exposition proves the corresponding sharp derivative norm for up through its exact dyadic derivative tiling [1]. The finite splines possess a stronger local plateau.

Theorem 4.3 (Finite-prefix derivative geometry). *Fix $m \geq 0$ and $n \geq m + 1$. Interpreting the top derivative at the limiting case $n = m + 1$ as its canonical almost-everywhere step representative, one has*

$$\|p_n^{(m)}\|_{L^\infty} = 2^{\binom{m+1}{2}}. \quad (30)$$

Moreover,

$$p_n^{(m)}(x) = (-1)^m 2^{\binom{m+1}{2}} \quad \text{for almost every } x \in [c_m - \delta_n, c_m + \delta_n]. \quad (31)$$

For $n \geq m + 2$ the derivative is classical and the equality holds pointwise on the closed interval. By evenness, the reflected plateau has the corresponding parity sign.

Proof. Factor

$$p_n = p_{m+1} * q_{m+1,n},$$

where $q_{m+1,n}$ is the convolution of the remaining boxes and is a probability density supported on

$$\left[-2^{-m-1} + \delta_n, 2^{-m-1} - \delta_n\right].$$

By [Equations \(17\) and \(18\)](#), the step function $p_{m+1}^{(m)}$ takes only the values 0 and $\pm 2^{\binom{m+1}{2}}$. Convolution with a probability measure cannot increase its supremum norm, proving the upper bound.

The point c_m is the midpoint of the rightmost nonzero cell of $p_{m+1}^{(m)}$, whose half-width is 2^{-m-1} . Translating the support of $q_{m+1,n}$ across any $x \in [c_m - \delta_n, c_m + \delta_n]$ remains inside that cell. Hence the convolution averages a constant and gives [Equation \(31\)](#), which also proves sharpness. $\square$

5 Sharp convergence of the ordinary partial products

5.1 Smooth-test and transport estimates

The coupling [Equation \(9\)](#) immediately gives, for $1 \leq p < \infty$,

$$W_p(\mathcal{L}(S_n), \mathcal{L}(X)) \leq \delta_n (\mathbb{E}|X|^p)^{1/p} \leq \delta_n. \quad (32)$$

The mean-zero tail improves smooth tests by one full power. If $\varphi \in C^2(\mathbb{R})$ has bounded second derivative, then

$$|\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)| \leq \frac{\delta_n^2}{18} \|\varphi''\|_\infty. \quad (33)$$

For convex φ , the left side has the nonnegative sign $\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)$ by conditional Jensen. The constant is sharp for $\varphi(x) = x^2$.

5.2 Exact C^r error law

Theorem 5.1 (Sharp ordinary-prefix law). *For every integer $r \geq 0$ and every $n \geq r + 3$,*

$$\boxed{\|p_n^{(r)} - \text{up}^{(r)}\|_\infty = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.} \quad (34)$$

The extremum is attained at

$$x = c_{r+2} = 1 - 2^{-(r+2)}, \quad (35)$$

and the signed value is

$$p_n^{(r)}(c_{r+2}) - \text{up}^{(r)}(c_{r+2}) = (-1)^{r+1} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (36)$$

If $\|f\|_{C^r} = \max_{0 \leq j \leq r} \|f^{(j)}\|_\infty$, then the same right-hand side is exactly $\|p_n - \text{up}\|_{C^r}$.

Proof. Let $g = p_n^{(r)}$. From [Equation \(10\)](#),

$$\text{up}^{(r)}(x) = \mathbb{E}g(x - \delta_n X).$$

The function g has a bounded second weak derivative when $n \geq r + 3$. Taylor's formula with integral remainder gives

$$g(x - h) = g(x) - hg'(x) + h^2 \int_0^1 (1-t)g''(x - th) dt.$$

Set $h = \delta_n X$ and take expectations. Since $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1/9$,

$$\begin{aligned} \left| \text{up}^{(r)}(x) - p_n^{(r)}(x) \right| &\leq \frac{\delta_n^2}{18} \left\| p_n^{(r+2)} \right\|_\infty \\ &= \frac{\delta_n^2}{18} 2^{\binom{r+3}{2}}, \end{aligned}$$

where [Theorem 4.3](#) was used in the second line. This is the claimed upper bound.

At $x = c_{r+2}$, the entire interval $[x - \delta_n, x + \delta_n]$ lies in the plateau from [Equation \(31\)](#) for $p_n^{(r+2)}$. The Taylor remainder therefore averages a constant, and every inequality above is an equality. Its sign is obtained from the plateau sign $(-1)^{r+2}$. Finally, the constants in [Equation \(34\)](#) increase strictly with the derivative order, proving the C^r statement. $\square$

For $r = 0$,

$$\|p_n - \text{up}\|_\infty = \frac{4}{9 \cdot 4^n} \quad (n \geq 3). \quad (37)$$

This is exactly one half of the repeated-integration error $8/(9 \cdot 4^n)$ in the frontier volume: omitting the tail has variance defect $-1/9$, whereas replacing it by a full-radius uniform box has variance defect $+2/9$.

5.3 Exact Kolmogorov distance

Let

$$P_n(x) = \int_{-\infty}^x p_n(t) dt, \quad P(x) = \int_{-\infty}^x \text{up}(t) dt. \quad (38)$$

Corollary 5.2 (Exact CDF convergence). *For every $n \geq 2$,*

$$\boxed{\|P_n - P\|_\infty = \frac{1}{9 \cdot 4^n}}. \quad (39)$$

The maximum is attained at $x = 1/2$, where $P_n(x) - P(x) = 1/(9 \cdot 4^n)$.

Proof. Apply the proof of [Theorem 5.1](#) to $g = P_n$. Now $g'' = p'_n$, whose supremum is 2 by [Theorem 4.3](#). The derivative p'_n is identically -2 on $[1/2 - \delta_n, 1/2 + \delta_n]$, so equality occurs at $1/2$. $\square$

5.4 L^p consequences and endpoint flatness

For $1 \leq p < \infty$ and $n \geq r + 3$,

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{L^p} \leq 2^{1/p} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (40)$$

The all-orders expansion in [Section 6](#) sharpens this to

$$4^n \left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{L^p} \longrightarrow \frac{1}{18} \left\| \text{up}^{(r+2)} \right\|_{L^p}, \quad 1 \leq p \leq \infty. \quad (41)$$

Although p_n vanishes on the thin endpoint layers $A_n < |x| \leq 1$, this causes no $O(\delta_n)$ uniform loss. Since every derivative of up vanishes at ± 1 , for every M and fixed r ,

$$\sup_{A_n \leq |x| \leq 1} |\text{up}^{(r)}(x)| = O_{r,M}(\delta_n^M). \quad (42)$$

The endpoint discrepancy is beyond every algebraic order; the 4^{-n} error is an interior moment effect.

6 All-orders differential expansion

6.1 The reciprocal product and Bell–Bernoulli coefficients

Near the origin, define

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{j=1}^{\infty} \alpha_j (2\pi z)^{2j}\right) = \sum_{m=0}^{\infty} a_m (2\pi z)^{2m}. \quad (43)$$

The logarithm of the sine product gives

$$\alpha_j = \frac{|B_{2j}|}{2j(2j)!(1-2^{-2j})}. \quad (44)$$

Thus

$$a_0 = 1, \quad ma_m = \sum_{j=1}^m j \alpha_j a_{m-j}, \quad (45)$$

and, in terms of the complete exponential Bell polynomial Y_m ,

$$a_m = \frac{1}{m!} Y_m(1!\alpha_1, 2!\alpha_2, \dots, m!\alpha_m). \quad (46)$$

The first coefficients are

$$\begin{aligned} a_0 &= 1, & a_1 &= \frac{1}{18}, & a_2 &= \frac{31}{16200}, \\ a_3 &= \frac{2347}{42865200}, & a_4 &= \frac{1904369}{1311675120000}, & a_5 &= \frac{3309760579}{88561680752160000}. \end{aligned} \quad (47)$$

6.2 Uniform expansion theorem

Theorem 6.1 (All-orders prefix expansion). *Fix integers $r \geq 0$ and $M \geq 1$. As $n \rightarrow \infty$,*

$$p_n^{(r)}(x) = \sum_{m=0}^{M-1} (-1)^m a_m \delta_n^{2m} \operatorname{up}^{(r+2m)}(x) + O_{r,M}(\delta_n^{2M}) \quad (48)$$

uniformly for $x \in \mathbb{R}$.

Proof. By Equation (10),

$$\Phi_n(\xi) = \Phi(\xi) \frac{1}{\Phi(\delta_n \xi)}.$$

Choose $0 < c < 1$. On $|\xi| \leq c/\delta_n$, the reciprocal is analytic and its Taylor remainder after $M-1$ terms is bounded by $C_M \delta_n^{2M} |\xi|^{2M}$. Multiplication by $|\xi|^r |\Phi(\xi)|$ is integrable because Φ is Schwartz, so Fourier inversion bounds the low-frequency remainder by $O_{r,M}(\delta_n^{2M})$.

For the high-frequency tail, retain any fixed K factors of Φ_n :

$$|\Phi_n(\xi)| \leq C_K |\xi|^{-K}.$$

Choosing $K > r + 2M + 1$ makes the integral over $|\xi| > c/\delta_n$ smaller than $O(\delta_n^{2M})$. The high-frequency tails of the finitely many $\Phi(\xi) \xi^{2m}$ terms are superalgebraically small. Finally,

$$\mathcal{F}^{-1}[(2\pi\xi)^{2m} \Phi(\xi)] = (-1)^m \operatorname{up}^{(2m)},$$

which yields Equation (48) after differentiating r times. $\square$

In particular,

$$p_n = \text{up} - \frac{\delta_n^2}{18} \text{up}'' + \frac{31\delta_n^4}{16200} \text{up}^{(4)} - \frac{2347\delta_n^6}{42865200} \text{up}^{(6)} + O(\delta_n^8). \quad (49)$$

The leading profile is therefore

$$4^n(p_n^{(r)} - \text{up}^{(r)}) \longrightarrow -\frac{1}{18} \text{up}^{(r+2)} \quad \text{uniformly.} \quad (50)$$

The sharp derivative norm of $\text{up}^{(r+2)}$ is exactly the constant in Equation (34), explaining why the asymptotic coefficient is already exact at the dyadic extremizer for every admissible n .

6.3 Signed geometric Richardson acceleration

Let $q = 1/4$ and, for $M \geq 1$, define

$$R_{n,M} = \sum_{j=0}^{M-1} w_{M,j} p_{n+j}, \quad (51)$$

where

$$\begin{aligned} w_{M,j} &= (-1)^{M-1-j} \frac{q^{\binom{M-j}{2}}}{(q; q)_j (q; q)_{M-1-j}} \\ &= \frac{(-1)^{M-1-j} q^{\binom{M-j}{2}}}{(q; q)_{M-1}} \begin{bmatrix} M-1 \\ j \end{bmatrix}_q. \end{aligned} \quad (52)$$

Here $(q; q)_k = \prod_{\ell=1}^k (1 - q^\ell)$. These are the Lagrange weights that extrapolate from the geometric nodes q^j , $0 \leq j < M$, to the origin:

$$\sum_{j=0}^{M-1} w_{M,j} q^{jm} = \delta_{m0} \quad (0 \leq m < M). \quad (53)$$

Consequently,

$$R_{n,M}^{(r)} = \text{up}^{(r)} - a_M q^{\binom{M}{2}} \delta_n^{2M} \text{up}^{(r+2M)} + O_{r,M}(\delta_n^{2M+2}). \quad (54)$$

The first weight vectors are

$$\begin{aligned} M=2: & \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \\ M=3: & \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \\ M=4: & \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \end{aligned}$$

Despite alternating signs, the coefficient variation stays uniformly bounded:

$$\sum_{j=0}^{M-1} |w_{M,j}| = \frac{(-q; q)_{M-1}}{(q; q)_{M-1}} < \frac{(-q; q)_\infty}{(q; q)_\infty} = 1.969260353668\dots \quad (55)$$

This supplies a stable signed arbitrary-order method. The next section constructs an entirely positive alternative.

7 A positive quadrature hierarchy for the exact tail

7.1 Up as a scale mixture of uniform boxes

The function up is even, nonnegative, and decreasing on $[0, 1]$. Define a measure η on $[0, 1]$ by

$$d\eta(r) = -2r \text{up}'(r) dr. \quad (56)$$

Integration by parts gives $\eta([0, 1]) = 1$.

Theorem 7.1 (Uniform scale mixture). *Let $R \sim \eta$ and $U \sim \text{Unif}[-1, 1]$ be independent. Then*

$$X \stackrel{d}{=} RU, \quad (57)$$

and

$$\text{up}(x) = \int_0^1 u_r(x) d\eta(r), \quad (58)$$

$$\Phi(\xi) = \int_0^1 \text{sinc}(2\pi r \xi) d\eta(r). \quad (59)$$

If $Z = R^2$ and ν is its law, then

$$d\nu(z) = -\text{up}'(\sqrt{z}) dz, \quad 0 < z < 1, \quad (60)$$

and

$$m_k := \int_0^1 z^k d\nu(z) = \mathbb{E}R^{2k} = (2k+1)\mathbb{E}X^{2k}. \quad (61)$$

Proof. For $x \geq 0$,

$$\int_0^1 u_r(x) d\eta(r) = \int_x^1 \frac{1}{2r} (-2r \text{up}'(r)) dr = \text{up}(x) - \text{up}(1) = \text{up}(x).$$

Evenness handles $x < 0$, proving the density identity and hence [Equation \(57\)](#). Fourier transformation gives [Equation \(59\)](#). The change of variables $z = r^2$ gives [Equation \(60\)](#). Finally,

$$\mathbb{E}R^{2k} = \int_0^1 -2r^{2k+1} \text{up}'(r) dr = 2(2k+1) \int_0^1 r^{2k} \text{up}(r) dr = (2k+1)\mathbb{E}X^{2k}.$$

□

The first moments of ν are

$$m_0 = 1, \quad m_1 = \frac{1}{3}, \quad m_2 = \frac{19}{135}, \quad m_3 = \frac{583}{8505}. \quad (62)$$

All m_k are rational. Therefore every monic orthogonal polynomial generated from ν has rational coefficients, and all Gaussian or Radau nodes are algebraic numbers obtainable by exact arithmetic before root isolation.

7.2 Quadrature tail surrogates

Let

$$Q = \sum_{j=1}^J w_j \delta_{z_j}, \quad 0 \leq z_j \leq 1, \quad w_j > 0, \quad \sum_j w_j = 1, \quad (63)$$

be a positive quadrature rule for ν . In spatial convolution formulas we use the measure convention $u_0 = \delta_0$. Let Z_Q have law Q , let U be an independent uniform variable, and define

$$Y_Q = \sqrt{Z_Q} U. \quad (64)$$

The associated positive approximant is

$$g_{n,Q} = p_n * \mathcal{L}(\delta_n Y_Q) = \sum_{j=1}^J w_j (p_n * u_{\delta_n \sqrt{z_j}}), \quad (65)$$

$$\hat{g}_{n,Q}(\xi) = \Phi_n(\xi) \sum_{j=1}^J w_j \operatorname{sinc}(2\pi \delta_n \sqrt{z_j} \xi). \quad (66)$$

Thus every rule produces a finite positive mixture of explicit piecewise-polynomial box splines.

For a smooth function h , introduce

$$A_h(z) = \frac{1}{2} \int_{-1}^1 h(\sqrt{z} u) du. \quad (67)$$

Repeated integration by parts gives the key transfer identity

$$A_h^{(\ell)}(z) = \frac{1}{2^{2\ell+1} \ell!} \int_{-1}^1 (1-u^2)^\ell h^{(2\ell)}(\sqrt{z} u) du. \quad (68)$$

Since

$$\int_{-1}^1 (1-u^2)^\ell du = \frac{2^{2\ell+1} (\ell!)^2}{(2\ell+1)!}, \quad (69)$$

we obtain

$$\|A_h^{(\ell)}\|_\infty \leq \frac{\ell!}{(2\ell+1)!} \|h^{(2\ell)}\|_\infty. \quad (70)$$

This is the bridge by which matching a moment of the squared scale Z cancels two spatial derivatives.

7.3 General exact quadrature error theorem

Assume Q is exact for polynomials of degree at most $\ell - 1$, and its Peano kernel of order ℓ has one sign. This includes Gaussian, one-endpoint Radau, and Lobatto rules for a positive measure on an interval [11, 12]. Set

$$\varepsilon_{Q,\ell} = Q(z^\ell) - m_\ell, \quad \Gamma_{Q,\ell} = \frac{|\varepsilon_{Q,\ell}|}{(2\ell+1)!}. \quad (71)$$

Theorem 7.2 (Exact positive quadrature law). *Suppose $\varepsilon_{Q,\ell} \neq 0$. For every $r \geq 0$ and $n \geq r + 2\ell + 1$,*

$$\left\| g_{n,Q}^{(r)} - \operatorname{up}^{(r)} \right\|_\infty = \Gamma_{Q,\ell} 2^{\binom{r+2\ell+1}{2}} 2^{-2\ell n}. \quad (72)$$

The extremum occurs at $x = c_{r+2\ell}$ and has sign $\operatorname{sgn}(\varepsilon_{Q,\ell})(-1)^r$.

Proof. Put $f = p_n^{(r)}$ and, for fixed x , let $h_x(v) = f(x - \delta_n v)$. Then

$$g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x) = Q(A_{h_x}) - \nu(A_{h_x}).$$

A one-signed Peano kernel and exactness through degree $\ell - 1$ imply

$$|Q(A) - \nu(A)| \leq \frac{|Q(z^\ell) - \nu(z^\ell)|}{\ell!} \|A^{(\ell)}\|_\infty.$$

Using Equation (70) and $\|h_x^{(2\ell)}\|_\infty = \delta_n^{2\ell} \|p_n^{(r+2\ell)}\|_\infty$ gives

$$|g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x)| \leq \Gamma_{Q,\ell} \delta_n^{2\ell} 2^{\binom{r+2\ell+1}{2}}.$$

At $x = c_{r+2\ell}$, every argument $x - \delta_n \sqrt{z} u$ lies in the constant plateau of $p_n^{(r+2\ell)}$ from Theorem 4.3. Hence $A_{h_x}^{(\ell)}$ is constant, the Peano inequality is an equality, and the asserted sign follows. $\square$

Two familiar approximations fit the same theorem. Ordinary truncation is $Q = \delta_0$, $\ell = 1$, with $\Gamma = 1/18$. The repeated-integration spline is $Q = \delta_1$, $\ell = 1$, with $\Gamma = 1/9$. Their factor-of-two error difference is therefore a one-line quadrature moment calculation.

7.4 Explicit positive accelerated schemes

Example 7.3 (One-node Gauss: a single variance-matched box). The one-node Gauss rule for ν is $Q = \delta_{1/3}$. It matches m_0, m_1 and has

$$\varepsilon_{Q,2} = \frac{1}{9} - \frac{19}{135} = -\frac{4}{135}, \quad \Gamma_{Q,2} = \frac{1}{4050}.$$

Thus

$$G_n = p_n * u_{\delta_n/\sqrt{3}} \tag{73}$$

is a single positive box spline, and for $n \geq r + 5$,

$$\|G_n^{(r)} - \text{up}^{(r)}\|_\infty = \frac{2^{\binom{r+5}{2}}}{4050 \cdot 16^n}. \tag{74}$$

Its support radius is $1 - \delta_n + \delta_n/\sqrt{3} < 1$. Endpoint flatness makes the support deficit superalgebraically small, while the interior error is exactly fourth order.

Example 7.4 (Two-node Lobatto: exact support and fourth order). The endpoint rule matching $m_1 = 1/3$ is

$$Q = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1. \tag{75}$$

It gives

$$H_n = \frac{2}{3}p_n + \frac{1}{3}(p_n * u_{\delta_n}), \tag{76}$$

which is nonnegative, integrates to one, and has exact support $[-1, 1]$. Since

$$\varepsilon_{Q,2} = \frac{1}{3} - \frac{19}{135} = \frac{26}{135}, \quad \Gamma_{Q,2} = \frac{13}{8100},$$

$$\|H_n^{(r)} - \text{up}^{(r)}\|_\infty = \frac{13 \cdot 2^{\binom{r+5}{2}}}{8100 \cdot 16^n} \quad (n \geq r + 5). \tag{77}$$

This explicitly realizes the positive, compactly supported, variance-matched 16^{-n} direction proposed but not constructed in the frontier document.

Example 7.5 (Two-node right Radau: exact support and sixth order). The two-node right-Radau rule, constrained to contain $z = 1$ and exact through z^2 , is

$$Q = \frac{15}{16}\delta_{13/45} + \frac{1}{16}\delta_1. \quad (78)$$

Its first missed moment is

$$\varepsilon_{Q,3} = \frac{704}{42525}, \quad \Gamma_{Q,3} = \frac{44}{13395375}. \quad (79)$$

Hence

$$J_n = \frac{15}{16}(p_n * u_{\delta_n \sqrt{13/45}}) + \frac{1}{16}(p_n * u_{\delta_n}) \quad (80)$$

is positive with exact support $[-1, 1]$, and

$$\left\| J_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{44 2^{\binom{r+7}{2}}}{13395375 64^n} \quad (n \geq r + 7). \quad (81)$$

7.5 Arbitrary order with positivity

Standard quadrature immediately produces a hierarchy.

Rule with M nodes	Exact degree	ℓ	Rate	Support behavior
Gauss	$2M - 1$	$2M$	2^{-4Mn}	Positive; strictly inside $[-1, 1]$
Right Radau (node 1)	$2M - 2$	$2M - 1$	$2^{-(4M-2)n}$	Positive; exact support $[-1, 1]$
Lobatto (nodes 0, 1, $M \geq 2$)	$2M - 3$	$2M - 2$	$2^{-(4M-4)n}$	Positive; exact support, includes p_n

The constants are not merely asymptotic: once $n \geq r + 2\ell + 1$, they are exactly

$$\frac{|Q(z^\ell) - m_\ell|}{(2\ell + 1)!} 2^{\binom{r+2\ell+1}{2}}. \quad (82)$$

Thus positive arbitrary-order acceleration reduces to the classical algebra of orthogonal polynomials for the explicitly defined measure $d\nu(z) = -\text{up}'(\sqrt{z}) dz$.

8 Numerical and exact-arithmetic experiments

8.1 Method

The accompanying script uses the exact product samples

$$\Phi_n(k/L), \quad \Phi(k/L),$$

on a periodic box of length $L = 4$ and applies an inverse FFT with 2^{17} modes. Every function considered is supported in $[-1, 1]$, so its periodization has no overlap. Derivatives are obtained by multiplying the transform by $(2\pi i \xi)^r$. The infinite product is truncated after 48 factors, beyond double-precision relevance on the active frequency range. Separate routines evaluate Equation (12) with exact rational arithmetic, generate the a_m , and generate the Richardson weights.

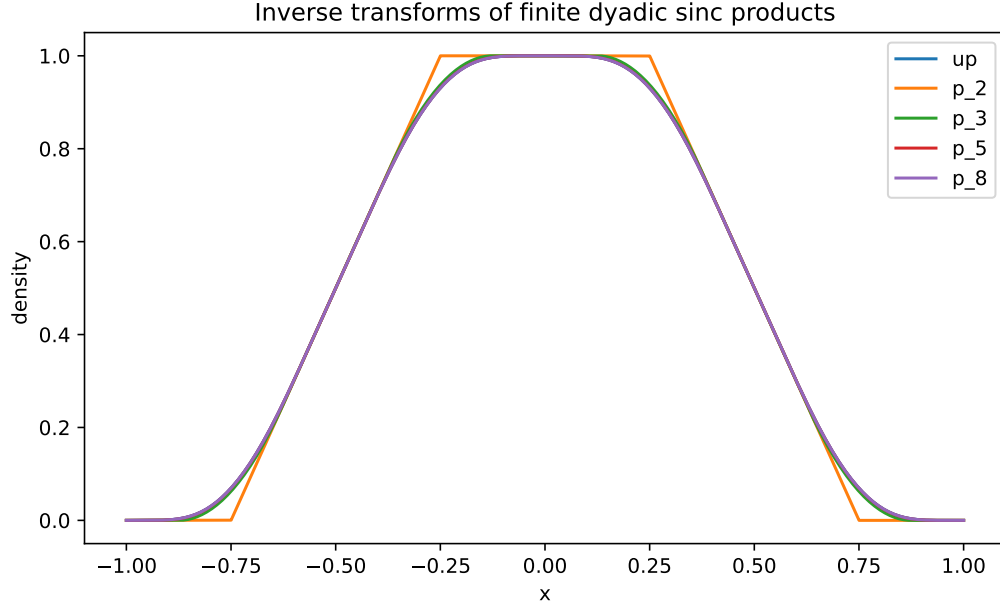


Figure 1: Inverse transforms of the first few finite sinc products. The support expands to $[-1, 1]$, the polynomial degree and smoothness increase, and the central plateau shrinks dyadically.

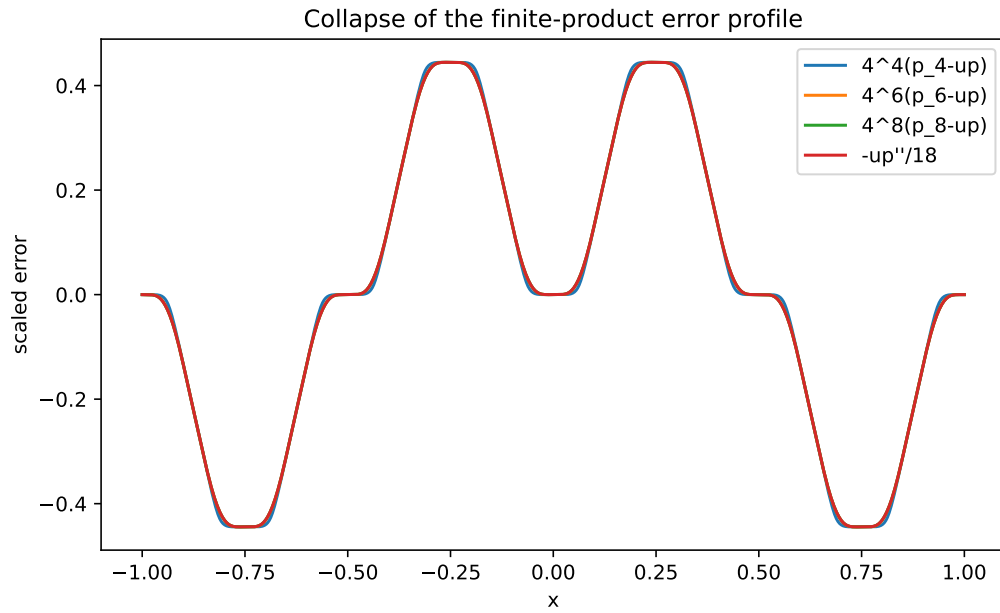


Figure 2: The profiles $4^n(p_n - \text{up})$ collapse onto $-\text{up}''/18$, in agreement with [Equation \(50\)](#).

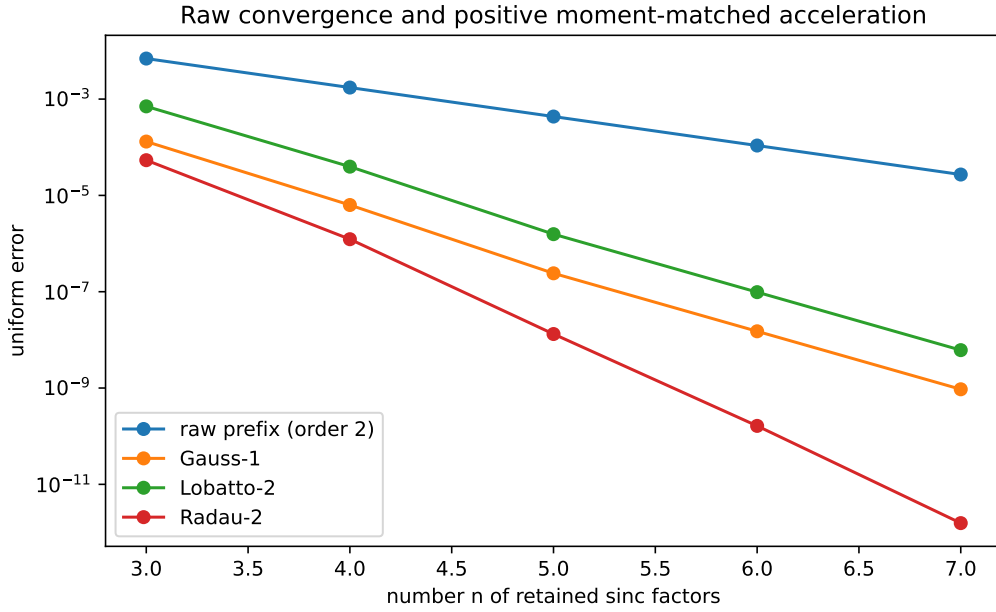


Figure 3: Raw second-order convergence and positive Gauss, Lobatto, and Radau tail acceleration. The highest-order Radau values approach the double-precision floor.

8.2 Verification of exact constants

Selected FFT ratios of measured supremum error to the exact formula are shown below.

Scheme	r	n	measured error	measured/exact
Raw prefix	0	3	$6.94444444448 \times 10^{-3}$	1.000000000006
Raw prefix	2	5	$5.55555555771 \times 10^{-2}$	1.000000000387
Raw prefix	4	7	7.11111128754	1.000000024810
Gauss-1	0	5	$2.41126543254 \times 10^{-7}$	1.000000000183
Lobatto-2	0	5	$1.56732253097 \times 10^{-6}$	1.000000000069
Radau-2	0	7	$1.56639950187 \times 10^{-12}$	1.000079775969

The raw and fourth-order laws agree essentially to machine precision. The Radau error is already 1.6×10^{-12} at its first admissible n , so Fourier truncation and roundoff visibly affect the last digits. Full tables are included in `sharp_error_verification.csv` and `positive_acceleration_verification.csv`.

9 Algorithms and implementation considerations

9.1 Exact cell construction

There are three complementary exact representations.

Global truncated powers. Equation (12) evaluates a point in $O(2^n)$ signed rational operations and is ideal for proof and low-order checks. Only knots left of x need be included.

Primitive recursion. If a piecewise-polynomial primitive of p_{n-1} is stored cellwise, then [Equation \(19\)](#) evaluates p_n by two shifted primitive values. This constructs the complete spline with exact coefficients while reusing adjacent-cell data.

Top-down integration of the Thue–Morse step. Start from the top derivative values in [Equations \(17\)](#) and [\(18\)](#), integrate cell by cell, and determine constants by continuity and vanishing at the left support endpoint. This uses only additions, multiplications by the mesh, and division by small integers. It is particularly attractive for formal verification.

9.2 Positive quadrature construction

The moment sequence $m_k = (2k + 1)\mathbb{E}X^{2k}$ can be generated exactly from the cumulants [Equation \(25\)](#). A practical M -node Gauss construction is:

1. compute $m_0, \dots, m_{2M}$ as rational numbers;
2. perform exact Gram–Schmidt on $1, z, \dots, z^M$, or solve the Hankel system for the monic orthogonal polynomial P_M ;
3. isolate the M simple roots of P_M in $(0, 1)$;
4. solve the Vandermonde moment equations for positive weights;
5. insert the nodes and weights in [Equation \(66\)](#) or [Equation \(65\)](#).

Right-Radau rules are obtained by orthogonalizing against the modified measure $(1 - z) d\nu(z)$ and then adjoining $z = 1$; Lobatto rules use $z(1 - z) d\nu(z)$ and adjoin both endpoints. Exact rational recurrence coefficients make certified algebraic output feasible.

9.3 Numerical stability

The global formula [Equation \(12\)](#) contains large alternating terms and should not be evaluated naively in floating point at high n . Stable options are:

- exact rational arithmetic followed by one final conversion;
- cellwise Horner evaluation after primitive recursion;
- FFT inversion when values on a dense uniform grid are needed;
- convolution on adaptive polynomial pieces for positive quadrature mixtures.

The Richardson coefficients have uniformly bounded ℓ^1 norm by [Equation \(55\)](#); their cancellation is therefore much milder than a generic high-order extrapolation. Positive quadrature schemes avoid cancellation entirely.

10 Further deductions and research directions

10.1 Results that appear immediately accessible

Problem 10.1 (Lean formalization of the prefix law). *Formalize [Theorems 3.1](#), [4.3](#) and [5.1](#). The proof ingredients are already close to the repository’s APIs: finite uniform convolutions, Thue–Morse prefix sums, the exact tail decomposition, bounded weak second derivatives, and the formally proved sharp derivative tiling of up.*

Problem 10.2 (Certified quadrature generator). *Build an exact-arithmetic pipeline that returns the rational orthogonal polynomial, certified algebraic nodes, and positive weights for Gauss and right-Radau rules of ν . The output would be a library of positive compactly supported approximants with machine-checkable error constants.*

Problem 10.3 (Norms beyond L^∞). *The expansion gives sharp leading constants in every L^p norm. The exact derivative distribution identities in the primary exposition suggest that some L^p constants, rearrangement-invariant norms, or signed error distributions may admit closed forms.*

10.2 Conjectures

Conjecture 10.4 (q -special orthogonal structure). *The monic orthogonal polynomials of $d\nu(z) = -\text{up}'(\sqrt{z})dz$ have recurrence coefficients admitting closed q -product or q -continued-fraction expressions at $q = 1/4$. Their rationality is automatic from the rational moments; the conjecture is that the rational numbers have a simple structural form involving $(q; q)_m$, Gaussian binomials, or the reciprocal-product coefficients a_m .*

Conjecture 10.5 (Lambert edge scaling of quadrature nodes). *The extreme Gauss and Radau nodes have refined endpoint asymptotics controlled by the same lower-Lambert phase that governs the flat endpoint of the Fabius/Rvachev function. The leading zero distribution may be classical, but the edge correction should retain the repository's log-periodic dyadic phase.*

Conjecture 10.6 (Positive minimax tail rule). *Among positive M -component uniform scale mixtures supported in $[-1, 1]$ and required to include the endpoint scale 1, the right-Radau rule minimizes the first nonzero worst-case C^r error constant simultaneously for all fixed r . The exact plateau formula reduces this to an extremal moment problem on $[0, 1]$.*

Conjecture 10.7 (Shape preservation). *For every right-Radau rule of ν and every sufficiently large n , the accelerated density $g_{n,Q}$ is strictly decreasing on $(0, 1)$ and has no spurious interior oscillation. Positivity is automatic, but monotonicity of a finite mixture is not; a proof may follow from total positivity of the uniform-box kernel.*

10.3 Broader extensions

Other geometric bases. For $0 < q < 1$, infinite products generated by widths $(1-q)q^j$ have the same exact head-tail scaling. The Bell-Bernoulli expansion and positive tail quadrature survive. The dyadic case $q = 1/2$ is exceptional because subset sums form a uniform grid and the top derivative is exactly Thue-Morse. Determining the sharp C^r constants for general q is a natural next problem.

Multivariate box splines. Products of directional sinc factors invert to multivariate box splines. A geometric infinite product produces a compactly supported C^∞ atomic function, and finite prefixes again provide polynomial approximants. Positive quadrature of the tail scale may extend with radial or matrix-valued moments.

Inverse Fabius propagation. On compact subintervals where up is bounded away from zero, the exact error expansion can be transported through inverse-function calculus to obtain high-order approximations of the inverse Fabius function. Near the flat endpoint, the algebraic prefix expansion must be matched with the Lambert- W asymptotic already developed in the repository.

Spectral and sampling questions. The zero multiplicities [Equation \(24\)](#) suggest dyadic Strang–Fix-type conditions, multiresolution reconstruction formulas, and exact cancellation identities for sampled derivatives. The finite splines could serve as compactly supported preconditioners whose approximation order is enhanced by the positive tail rules.

11 Conclusion

The inverse Fourier transforms of finite dyadic sinc products are much more rigid than generic finite convolutions. Binary subset sums force a uniform knot grid; Thue–Morse signs govern the top derivative; and a precise derivative plateau survives every later convolution. Combined with the exact self-similar tail of the Rvachev distribution, this local geometry turns a Taylor bound into an equality and yields the sharp law

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The same mechanism converts one-signed quadrature errors into exact functional errors. The resulting Gauss–Radau–Lobatto hierarchy supplies positive piecewise-polynomial approximants of arbitrary order, with explicit constants and exact support whenever a right endpoint node is included. This closes the repository’s variance-matched 16^{-n} frontier and opens a tractable new interface among dyadic harmonic analysis, Thue–Morse combinatorics, Bell–Bernoulli series, orthogonal polynomials, and certified numerical approximation.

A Proof of the scale-derivative identity

For completeness, we prove [Equation \(68\)](#). The case $\ell = 0$ is the definition. Assume it holds at order ℓ . Differentiating under the integral and using $\partial_z h^{(2\ell)}(\sqrt{z}u) = u h^{(2\ell+1)}(\sqrt{z}u)/(2\sqrt{z})$ gives

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+2}\ell!\sqrt{z}} \int_{-1}^1 u(1-u^2)^\ell h^{(2\ell+1)}(\sqrt{z}u) du.$$

Since

$$\frac{d}{du}(1-u^2)^{\ell+1} = -2(\ell+1)u(1-u^2)^\ell,$$

integration by parts has no boundary term and yields

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+3}(\ell+1)!} \int_{-1}^1 (1-u^2)^{\ell+1} h^{(2\ell+2)}(\sqrt{z}u) du.$$

This is the desired induction step. The formula extends continuously to $z = 0$ by Taylor expansion.

B Reproducibility manifest

The archive accompanying this report contains:

- `finite_sinc_products_report.tex` – the complete source;
- `finite_sinc_products_report.pdf` – the rendered report;
- `finite_sinc_experiments.py` – documented numerical and exact code;
- `finite_sinc_approximants.pdf` – generated approximation figure;

- `scaled_error_profiles.pdf` – generated error-profile figure;
- `convergence_comparison.pdf` – generated convergence figure;
- `sharp_error_verification.csv` – raw-prefix numerical table;
- `positive_acceleration_verification.csv` – accelerated numerical table;
- `exact_coefficients.txt` – exact series and Richardson coefficients;
- `exact_rational_samples.csv` – exact rational spline samples.

Run

```
python finite_sinc_experiments.py --output-dir . --fft-power 17
```

to regenerate every figure and table.

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